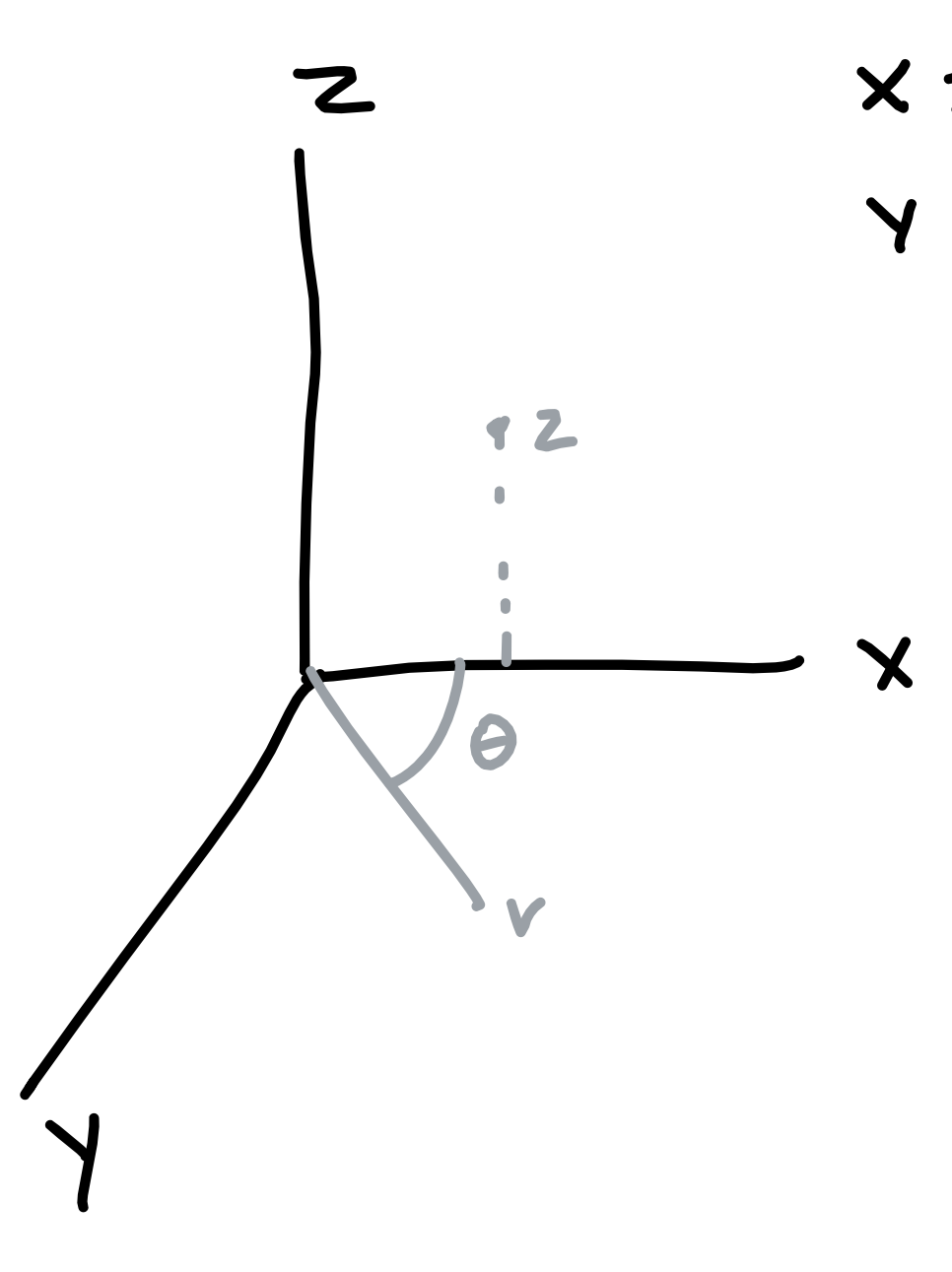
Cylindrical and Spherical Coordinates

section 11.7
9-3-26

A) Cylindrical



$x = r\cos\theta$

$y = r\sin\theta$

$x^2+y^2 = r^2$

$\tan\theta = \frac{y}{x}$

ex) convert $P(1, \sqrt{3}, 2)$
$\quad\quad\quad\quad x \quad y \quad z$

$r^2 = x^2+y^2 \quad\quad \tan\theta = \frac{y}{x}$

$6^2 = 4 \quad\quad \tan\theta = \frac{\sqrt{3}}{1}$

ex) Convert $y^2 = x$

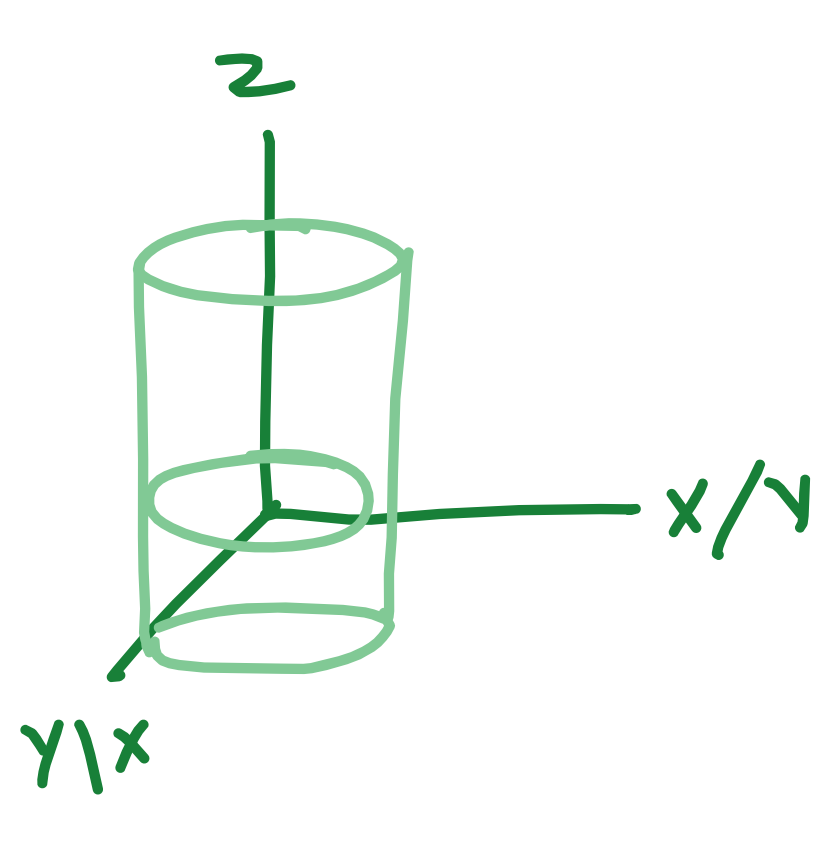
$(r\sin\theta)^2 = r\cos\theta$

$r^2\sin^2\theta = r\cos\theta$

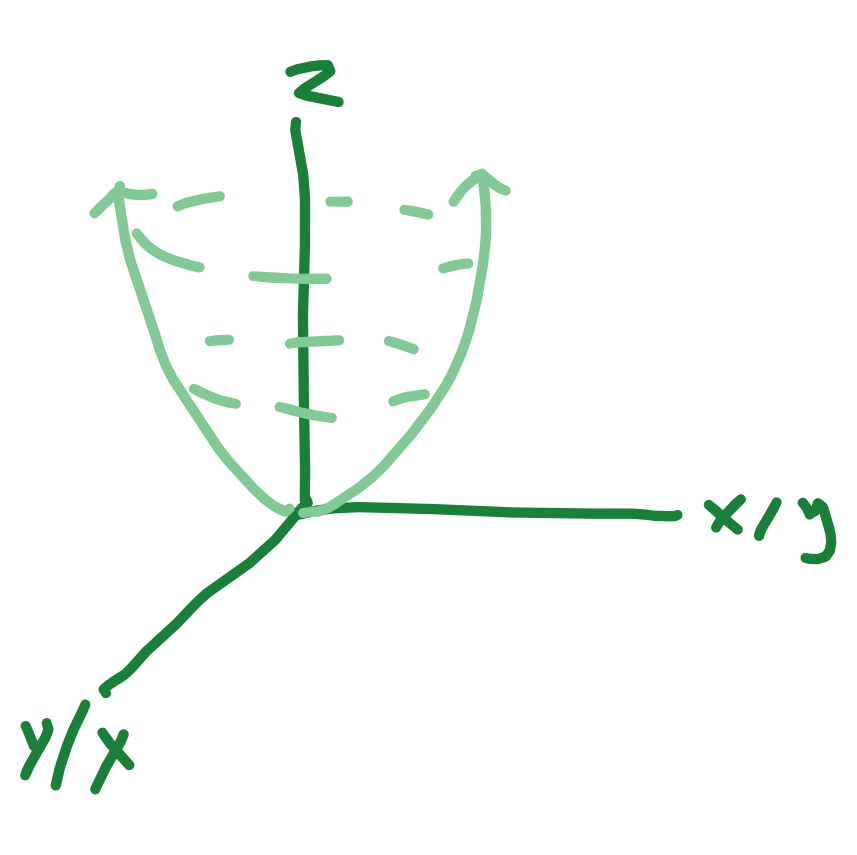
$r = \frac{\cos\theta}{\sin^2\theta} = \cot\theta \cdot \cos\theta$

ex) use when rotate on z-axis

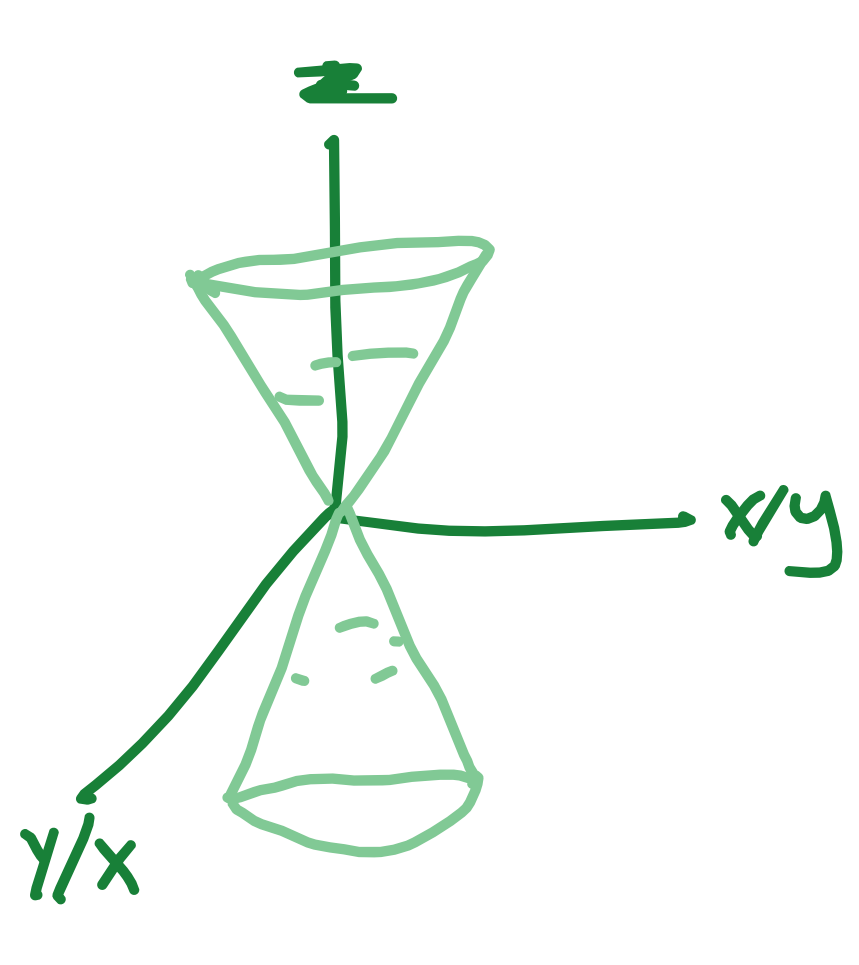
1) $x^2+y^2=9$



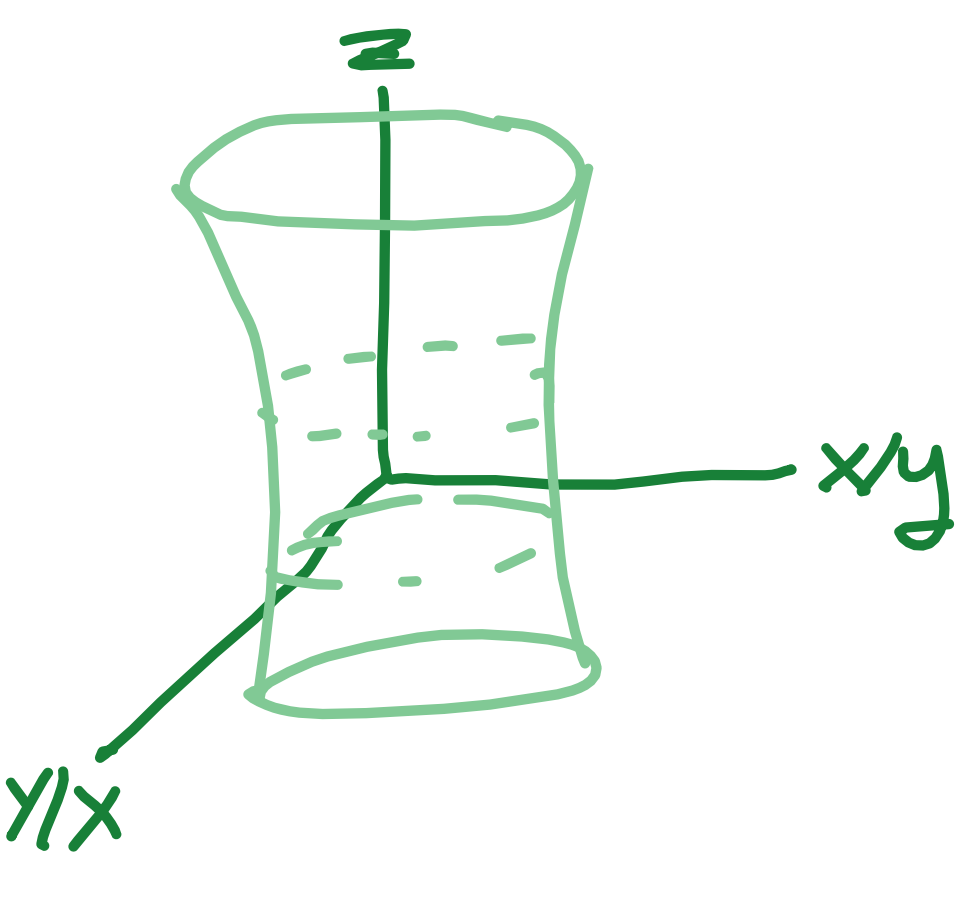
2) $x^2+y^2 = 4z$



3) $x^2+y^2=z^2$



4) $x^2+y^2-z^2=1$



HW 11.7 9/4

5) $x = 4\cos\left(\frac{7\pi}{6}\right) = -2\sqrt{3}$
$y = 4\sin\left(\frac{7\pi}{6}\right) = -2$
$z = 3$
$(-2\sqrt{3}, -2, 3)$

9) $r = \sqrt{2^2+(-2)^2} = 2\sqrt{2}$
$\theta = \arctan(-1) = -\frac{\pi}{4}$
$z = -4$
$(2\sqrt{2}, -\pi/4, -4)$

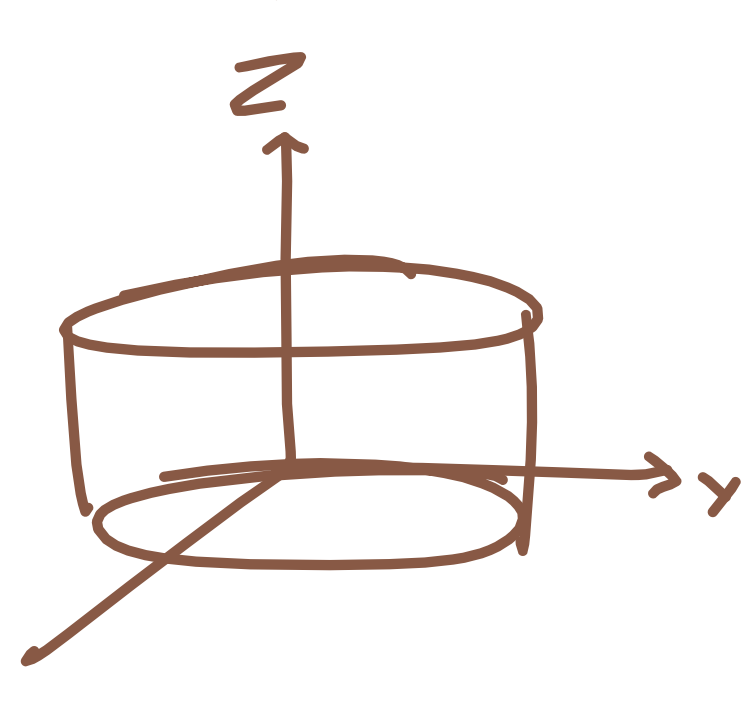
13)
$z = 4$ (cylindrical)

15)
$x^2+y^2=r^2$ so $r^2+z^2=17$ (cylindrical)

17)
$r = \sec\theta \cdot \tan\theta$ (cylindrical)

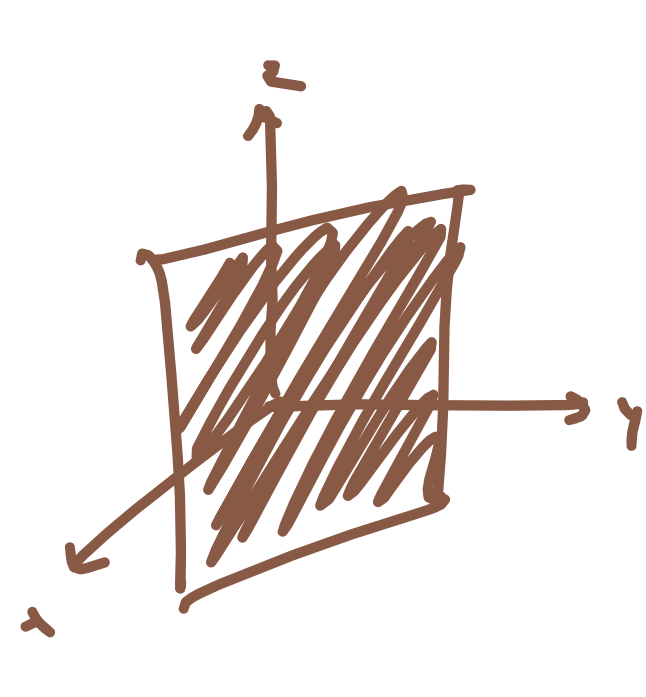
19)
$r^2\sin^2\theta + z^2 = 10$ (cylindrical)

21)
$r = 3$
$\sqrt{x^2+y^2} = 3$
$x^2+y^2=9$

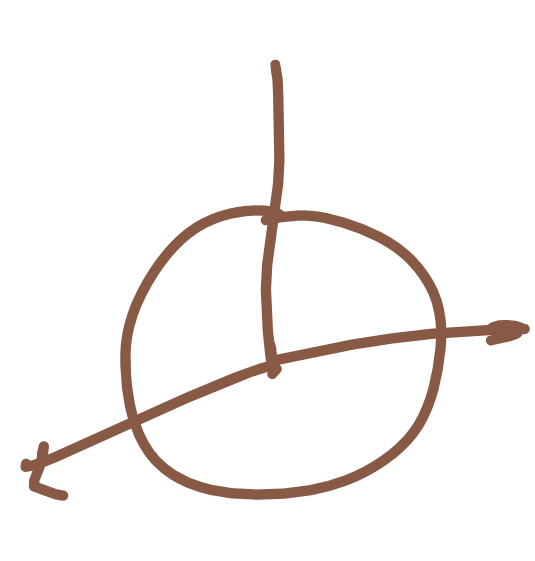


23)
$\theta = \pi/6$
$\tan \pi/6 = y/x$
$1/\sqrt{3} = y/x$

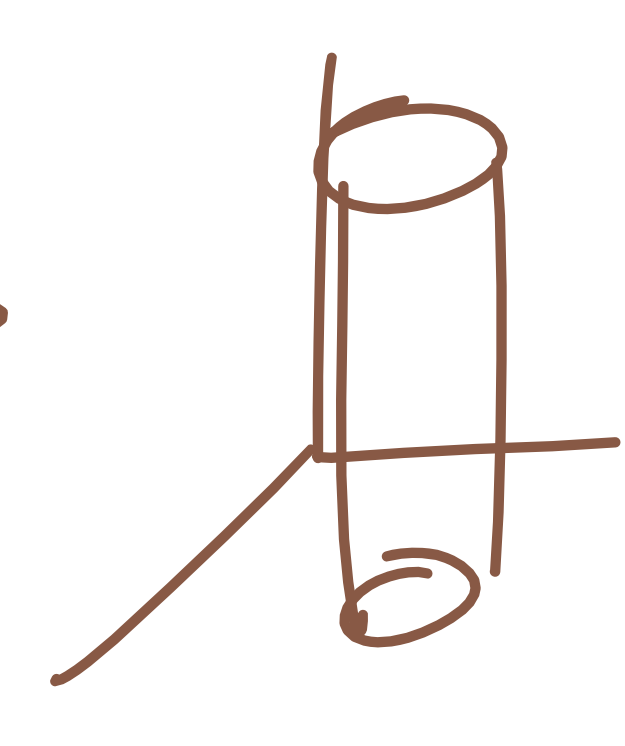
$x = \sqrt{3}y$
$x - \sqrt{3}y = 0$



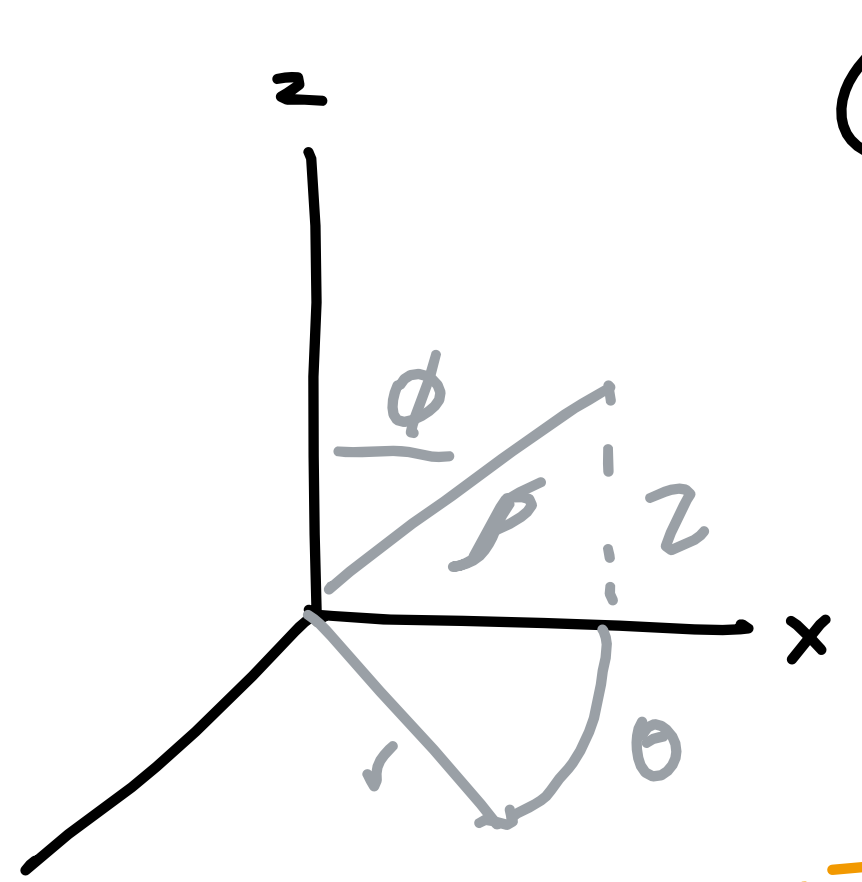
25)
$r^2+z^2=5$
$x^2+y^2+z^2=5$



27)
$r = 2\sin\theta \quad r^2 = 2r\sin\theta$
$x^2+y^2 = 2y$
x^2+y^2-2y -->
$x^2+(y-1)^2 = 1$



B) Spherical



(ρ, θ, ϕ)
rho theta phi

$\tan\theta = y/x$
$x = r\cos\theta$
$y = r\sin\theta$

$\rho\sin\phi = r$

$\sin\phi = \frac{r}{\rho}$
$\cos\phi = \frac{z}{\rho}$
$\tan\phi = \frac{r}{z}$

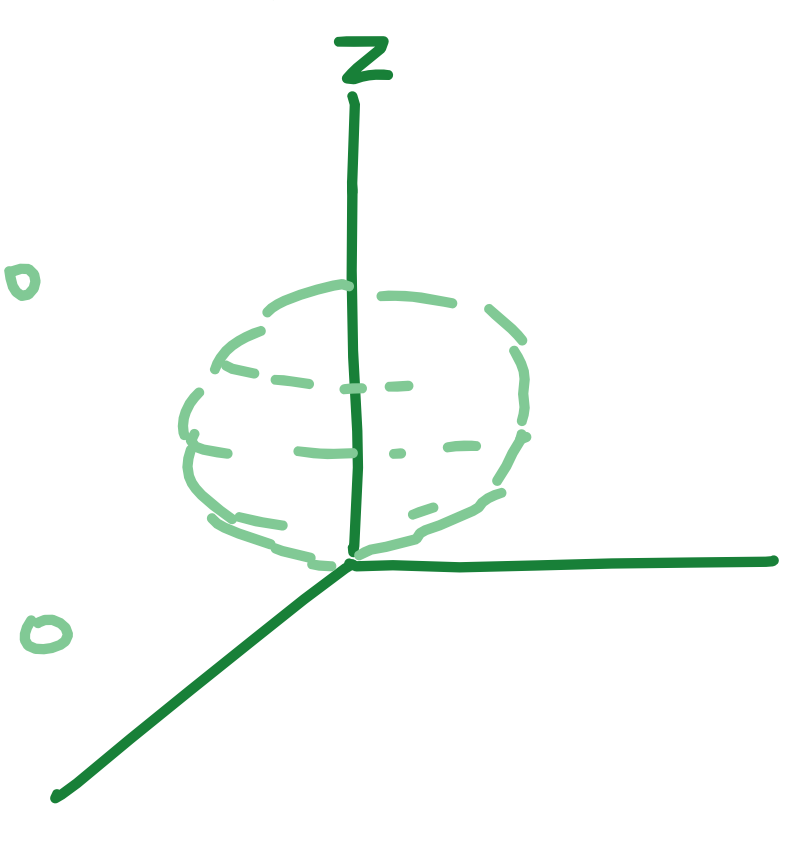
$x = \rho\sin\phi\cos\theta$
$y = \rho\sin\phi\cos\theta$
$z = \rho\cos\theta$

$\rho^2\sin^2\phi + \rho^2\cos^2\phi = \rho^2$

ex) $x^2+y^2+z^2-4z=0$

cyl. $r^2+z^2-4z=0$

sph $\rho^2 - 4\rho\cos\phi = 0$
$\rho = 4\cos\phi$



spherical ⟷ cylindrical

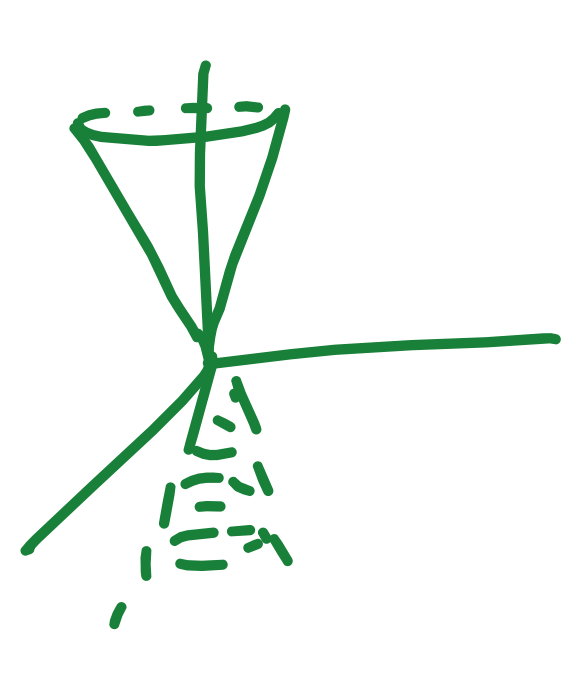
$(\rho, \theta, \phi) \quad\quad (r, \theta, z)$

$z = \rho\cos\phi \quad\quad r^2+z^2 = \rho^2$

$r^2 = \rho^2\sin^2\phi \quad\quad \cos\phi = \frac{z}{\sqrt{r^2+z^2}}$

ex) rect ⟶ spherical

$r^2 = z^2$
$r = z$
$\rho\sin\phi = \rho\cos\phi$
$\tan\phi = 1$
$\phi = \frac{\pi}{4}$



HW 11.7 9/8

31) $(4\sqrt{2}, \frac{2\pi}{3}, \frac{\pi}{4})$
$\quad\quad \rho \quad\quad \theta \quad\quad \phi$

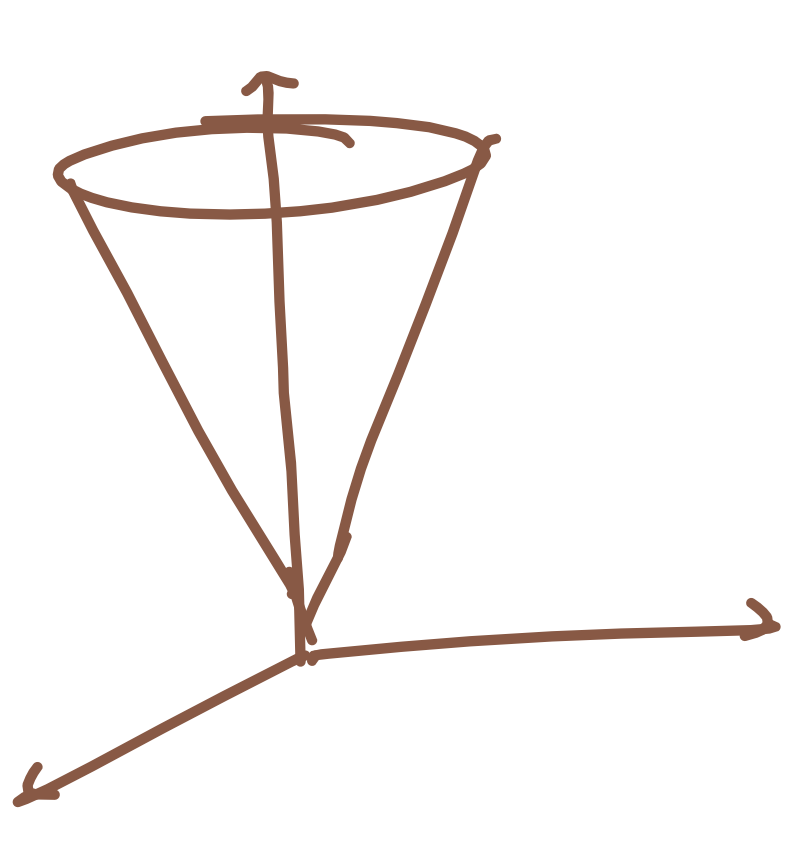
37) $(0, 0, 12)$
$\quad\quad x \quad y \quad z$

43) $x^2+y^2+z^2 = 49$
$\rho^2 = 49$
$\rho = 7$

45) $x^2+y^2 = 16$

51) $\phi = \frac{\pi}{6}$
$\cos\phi = \frac{z}{\sqrt{x^2+y^2+z^2}}$
$\frac{\sqrt{3}}{2} = \frac{z}{\sqrt{x^2+y^2+z^2}}$
$3/4 = \frac{z^2}{x^2+y^2+z^2}$

$3x^2+3y^2-z^2=0, \quad z \geq 0$



53) $\rho = 4\cos\phi$
$\sqrt{x^2+y^2+z^2} = \frac{4}{\sqrt{x^2+y^2+z^2}}$

$x^2+y^2+z^2-4z=0$
$x^2+y^2+(z-2)^2 = 4$

57) d
58) e
59) c
60) a
61) f
62) b

65) $(4, \frac{\pi}{2}, 4)$
$\rho = \sqrt{4^2+4^2} = 4\sqrt{2}$
$\theta = \frac{\pi}{2} \quad \phi = \arccos\left(\frac{4}{4\sqrt{2}}\right) = \frac{\pi}{4}$

$(4\sqrt{2}, \frac{\pi}{2}, \frac{\pi}{4})$ (spherical)

73) $(36, \pi, \frac{\pi}{2})$ (spherical)
$r = \rho\sin\phi = 36\sin\frac{\pi}{2} = 36$
$\theta = \pi$
$z = \rho\cos\phi = 36\cos\frac{\pi}{2} = 0$
$(36, \pi, 0)$ (cylindrical)

85) $x^2+y^2+z^2-2z=0$
a) $r^2+z^2-2z=0$
$r^2+(z-1)^2=1$
b) $\rho^2-2\rho\cos\phi=0$
$\rho(\rho-2\cos\phi)=0$
$\rho = 2\cos\phi$